ISPRS Journal of Photogrammetry and Remote Sensing 212 (2024) 422–439

Contents lists available at ScienceDirect

ISPRS Journal of Photogrammetry and Remote Sensing

journal homepage: www.elsevier.com/locate/isprsjprs

EarthVQANet: Multi-task visual question answering for remote sensing image understanding

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ARTICLE INFO

Keywords:
Visual question answering
Semantic segmentation
Multi-modal fusion
Multi-task learning
Knowledge reasoning

ABSTRACT

Monitoring and managing Earth's surface resources is critical to human settlements, encompassing essential tasks such as city planning, disaster assessment, etc. To accurately recognize the categories and locations of geographical objects and reason about their spatial or semantic relations , we propose a multi-task framework named EarthVQANet, which jointly addresses segmentation and visual question answering (VQA) tasks. EarthVQANet contains a hierarchical pyramid network for segmentation and semantic-guided attention for VQA, in which the segmentation network aims to generate pixel-level visual features and high-level object semantics, and semantic-guided attention performs effective interactions between visual features and language features for relational modeling. For accurate relational reasoning, we design an adaptive numerical loss that incorporates distance sensitivity for counting questions and mines hard-easy samples for classification questions, balancing the optimization. Experimental results on the EarthVQA dataset (city planning for Wuhan, Changzhou, and Nanjing in China), RSVQA dataset (basic statistics for general objects), and FloodNet dataset (disaster assessment for Texas in America attacked by Hurricane Harvey) show that EarthVQANet surpasses 11 general and remote sensing VQA methods. EarthVQANet simultaneously achieves segmentation and reasoning, providing a solid benchmark for various remote sensing applications. Data is available at http://rsidea.whu.edu.cn/EarthVQA.htm

1. Introduction

High-spatial resolution (HSR) remote sensing images provide rich details about object structures, facilitating macro-level observations of human settlements and ecosystems (Wang et al., 2022b). Large-scale HSR images often depict various ground objects with diverse spatial and spectral patterns, which can be highly complex and heterogeneous (Wieland et al., 2023). To characterize land-cover semantics from HSR image, intelligent interpretation of HSR images has been studied over the past few decades (Zhang et al., 2020b; Dimitrovski et al., 2023). Traditional methods for recognizing land-cover objects in HSR images rely on handcrafted visual characteristics such as object indices (Chen et al., 2004), filter responses (Pelletier et al., 2016), and object image analysis (Hossain and Chen, 2022). However, these methods have limited performance due to their reliance on expert experience and shallow features that cannot adequately capture complex urban scenes (Xiao et al., 2023). In recent years, data-driven methods represented by convolutional neural networks (CNNs) have gradually replaced handcrafted methods, offering powerful representation abilities thanks to the end-to-end hierarchical learning of deep semantic features (Martins et al., 2020). CNNs and their variants have advanced the development of HSR interpretation, particularly in the areas of object detection (Kellenberger et al., 2018), road extraction (Chen et al., 2021), and scene classification (Carbonneau et al., 2020).

To further advance comprehensive understanding, we construct the multi-level understanding process for HSR image (Fig. 1). #1 stage denotes the object extraction, which provides intuitive mapping results for users. This is implemented by a segmentation task and can also be replaced by object detection. #2 stage involves relational reasoning for interested objects, which provides comprehensive knowledge. There already exist many mature solutions (Zhao et al., 2022) to accurately extract the locations and categories of land-cover objects from HSR images. However, these can only provide land-cover maps and cannot achieve object-relational reasoning for #2 stage.

Visual question answering (VQA) (Antol et al., 2015) is a task aimed at answering customized questions by searching for visual clues in the

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https://doi.org/10.1016/j.isprsjprs.2024.05.001
Received 5 August 2023; Received in revised form 9 April 2024; Accepted 3 May 2024
Available online 18 May 2024
0924-2716/© 2024 Published by Elsevier B.V. on behalf of International Society for Photogrammetry and Remote Sensing, Inc. (ISPRS).

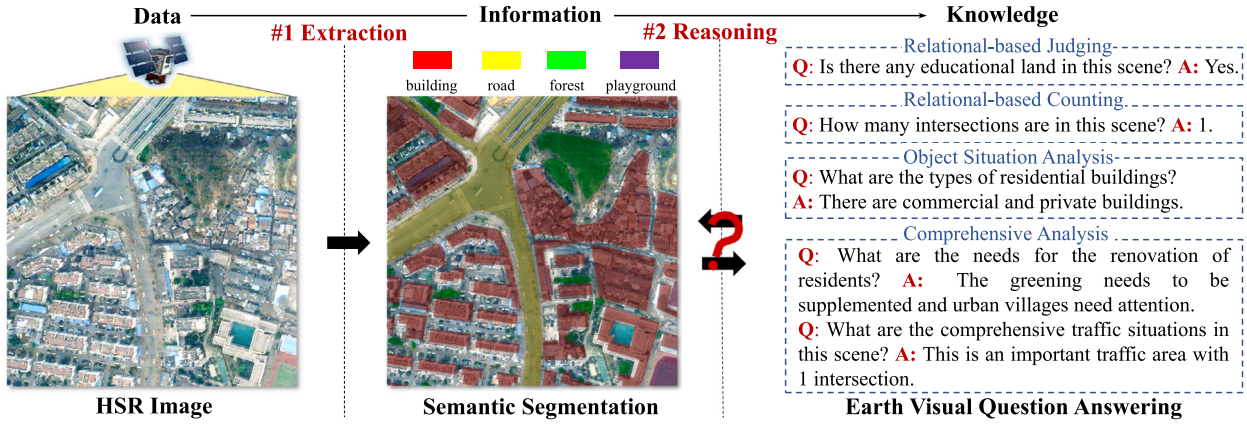


Fig. 1. Multi-level understanding process for remote sensing image. #1 stage performs information extraction of object categories and locations and #2 stage involves relational reasoning for comprehensive knowledge.

provided image. The properties of tasks are determined by the questions asked, and various analysis processes can be embedded according to specific applications (Abdelnour et al., 2023). VQA bridges the gap between vision and language, and plays a crucial role in assisting humans to understand images, such as blind reading (Tu et al., 2021), robot conversation (Gao et al., 2021a), etc.

There already exists preliminary research for remote sensing VQA (Lobry et al., 2020; Zheng et al., 2021), and most methods directly fuse the global image and language features. However, the global fusion mechanism cannot capture the relations for small-scale objects and fails to achieve accurate relations between multiple objects. Many computer vision VQA studies (Tan and Bansal, 2019; Anderson et al., 2018) show that local object features are very important for relational reasoning. Especially for HSR images, multiple objects with varied scales require more refined analysis.

To this end, we implement #2 stage by a novel semantic-guided VQA task. In this paper, we propose a multi-task framework, called EarthVQANet, to seamlessly combine the segmentation and VQA architectures for simultaneous learning. Guided by refined object semantics from the segmentation network, EarthVQANet can pay attention to each individual land-cover object and even its internal features, achieving complex reasoning. Main contributions of this paper are:

(1) **Task-Seg: Hierarchical pyramid segmentation network for visual features.** As information extraction serves as a foundation for VQA, a hierarchical pyramid network is designed to obtain accurate land-cover object semantics. The segmentation network consists of a scalable encoder, a pyramid pooling module and a pyramid decoder. The encoder shares the universal semantics for multi-tasks and is compatible with both CNN and Transformer architectures. The pyramid pooling module and pyramid decoder hierarchically capture multi-scale features for segmentation-specific representation. Overall, the hierarchical pyramid network provides visual features and pseudo masks with object semantics for VQA.

(2) **Task-VQA: Semantic-guided attention mechanism for relational reasoning.** The semantic-guided attention shares the universal semantics with segmentation and utilizes pseudo masks as semantic guidance. The semantic guidance spatially bootstraps semantics in a hierarchical structure. The semantic-guided self-attention firstly enhances visual features with object semantics, reasoning object external relations. The semantic-guided cross-attention then performs multi-modal interactions, aggregating visual clues via keywords in questions. The bidirectional mechanism is constructed to fully fuse visual and language features, summarizing knowledge according to required questions.

(3) **Adaptive numerical loss for balanced optimization.** Faced with questions with diverse difficulties and imbalanced answers, we design an adaptive numerical loss for balanced optimization. Specifically, the numerical difference penalty is added to regression questions and hard example mining is adopted for classification questions, jointly revising the optimization directions. Enhanced by these two strategies, adaptive numerical loss is finally combined with segmentation loss for multi-task learning.

The proposed EarthVQANet has demonstrated remarkable superiority in comprehensive tasks, providing not only accurate semantic maps but also interactive knowledge reasoning answers. EarthVQANet has achieved superior results on the EarthVQA dataset, RSVQA dataset and has also been successfully applied to disaster assessment using the FloodNet dataset. The rest of this paper is organized as follows. Section 2 introduces the related work for VQA. Sections 3 and 4 describe the details of EarthVQANet and EarthVQA dataset. Experimental results are analyzed in Sections 5 and 6. Section 7 concludes the paper.

2. Related work

2.1. Land-cover semantic segmentation

Remote sensing semantics segmentation aims to determine the given image with specific categories at a pixel level. Fully convolutional network (FCN), as a well-established architecture, has already achieved promising results in HSR segmentation tasks (Liu et al., 2023). Considering the rich details in HSR images, ResUNet (Diakogiannis et al., 2020) was designed with residual connections, atrous convolutions, and pyramid scene parsing pooling. These advanced modules contribute to multi-scale object recognition in land-cover mapping tasks. To preserve the object integrity, OCNN (Martins et al., 2020) integrates deep learning features with traditional object image analysis, reducing semantic inconsistency results. To further achieve large-scale mapping, LoveCS (Wang et al., 2022b) explores model transferability between different sensors. With the abandonment of inductive bias, various Transformer architectures (Wang et al., 2022a) have been introduced into HSR segmentation. UNetFormer (Wang et al., 2022c) designs an efficient global–local attention mechanism in the decoder, and achieves a good trade-off between accuracy and efficiency. Due to the excellent generalization ability, Swin-Transformer (Liu et al., 2021) has shown great potential in many competitions, such as landslide detection (Ghorbanzadeh et al., 2022), land-cover mapping (Hänsch et al., 2022), etc. These advanced segmentation methods serve as a solid benchmark for information extraction. EarthVQANet exhibits scalability and compatibility with both FCN and Transformer architectures.

Table 1
Comparison between EarthVQA and existing remote sensing VQA datasets.

Datasets	Source	Image size	Resolution (m)	#QA pairs	Bas Ju	Bas Co	Rel Ju	Rel Co	Obj An	Com An	Sem Mask	Land-Use
RSVQA-LR (Lobry et al., 2020)	Auto	256	10	77K	✓	✓	✓	✓	×	×	×	×
RSVQA-HR (Lobry et al., 2020)	Auto	512	0.15	955K	✓	✓	✓	✓	×	×	×	×
RSVQAxBen (Lobry et al., 2021)	Auto	120	10~60	15M	✓	×	✓	✓	×	×	×	✓
RSIVQA (Zheng et al., 2021)	Semi	512~4000	0.3~8	111K	✓	✓	✓	×	✓	×	×	×
HRVQA (Li et al., 2023b)	Semi	1024	0.08	1070K	✓	✓	×	✓	✓	×	×	×
CDVQA (Yuan et al., 2022b)	Auto	512	0.5~3	122K	✓	✓	×	×	×	×	✓	×
FloodNet (Rahnemoonfar et al., 2021)	Auto	3000–4000	0.015	11K	×	✓	✓	✓	✓	×	✓	×
TextRS-VQA (Bashmal et al., 2023)	Manual	256	0.06–5	6245	✓	✓	✓	✓	×	×	×	×
EarthVQA	Semi	1024	0.3	208K	✓	✓	✓	✓	✓	✓	✓	✓

The abbreviations are: Bas Ju (Basic Judging), Rel Ju (Relational-based Judging), Bas Co (Basic Counting), Rel Co (Relational-based Counting), Obj An (Object Situation Analysis), Com An (Comprehensive Analysis), Sem Mask (Semantic Mask).

2.2. Visual question answering

Early research consider VQA as the fusion of global image and language features (Antol et al., 2015). The image and language features are individually processed by CNN and Recurrent Neural Network (RNN), and multimodal global features are fused to predict the final answer. Stacked attention network (SAN) (Yang et al., 2016) designs multiple attention layers to locate the visual clues layer by layer. In order to reason complex relations efficiently, Bottom-Up-Top-Down (BUTD) (Anderson et al., 2018) has used Faster-RCNN features to introduce object features. The image regions function as a restricted attention mechanism as the fusion model can efficiently capture key objects. Pythia (Jiang et al., 2018) further re-implemented the BUTD model and ensemble diverse models trained with different settings, achieving better performances. Cubic visual attention (Song et al., 2018) has applied a channel and spatial attention on object regions to enhance the associated convolutional features. By imitating the human mislabeling behavior, semantic noisy label correction (Zhang et al., 2023a) has improved the robustness of VQA under noisy labels. VILBERT (Lu et al., 2019) designs a two-stream BERT to process both visual and textual inputs in separate streams that interact through co-attentional transformer layers. To simplify and unify, VL-BERT (Su et al., 2020) regard the visual and language features as inputs for one BERT model without any restriction on the attention patterns, which facilitates the interaction of features early and freely. Modular Co-Attention Network (MCAN) (Yu et al., 2019) further implements the fusion module with Transformer to interact the visual and language features. D-VQA (Wen et al., 2021a) integrates a unimodal bias detection module, thereby efficaciously alleviating negative biases. With the advent of large multi-modal models, the conditional generative models (i.e., BLIP-2 Li et al., 2023a, Instruct-BLIP Dai et al., 2024) also show promising results on generic VQA tasks. These large multi-modal methods can be fine-tuned by injection of a few learnable parameters when applied to remote sensing VQA tasks. Recently, many advanced VQA algorithms (Lin et al., 2022; Gao et al., 2022; Zhang et al., 2020a) introduce the external knowledge database, i.e., Wikidata, to improve the generalizability. Besides, various research (Gao et al., 2021b; Zeng et al., 2022) has applied VQA from single-frame images to video interactions.

In the remote sensing community, there has been preliminary research on VQA. As for datasets, RSVQA-LR (Lobry et al., 2020) and RSVQA-HR (Lobry et al., 2020) constructed QA pairs from Open Street Map (OSM) properties following the handcrafted rules. RSVQAxBen (Lobry et al., 2021) was built on the 2018 CORINE Land Cover database, querying land-cover types at different levels. RSIVQA dataset (Zheng et al., 2021) collects images from existing classification and object detection datasets (AID (Xia et al., 2017), HRRSD (Zhang et al., 2019), etc.) and automatically generates answers from their semantic labels. In addition, some handcrafted QA pairs were added for complexity. Similar to the RSIVQA dataset, the TextRS-VQA dataset (Bashmal et al., 2023) also collects scene classification images from AID (Xia et al., 2017), PatternNet (Zhou et al., 2018), UC Merced (Yang and Newsam, 2010), and NWPU45 (Cheng et al., 2017) datasets. The QA pairs are manually designed to ensure balanced questions. Similarly, the HRVQA (Li et al., 2023b) dataset constructed automatic QA pairs from PDOK open-data source and manually annotated some relational reasoning-based samples. CDVQA (Yuan et al., 2022b) generates the QA pairs for change details based on the SECOND (Yang et al., 2021) change detection dataset. FloodNet (Rahnemoonfar et al., 2021) focuses on the Harvey Hurricane event, and designs disaster assessment QA pairs. As shown in Table 1, EarthVQA dataset involves more complex and practical questions to meet the needs of city planning requirements.

Most remote sensing VQA methods (Lobry et al., 2020; Rahnemoonfar et al., 2021) directly predict answers from the raw images, ignoring the important information extraction stage. Prompt-RSVQA (Chappuis et al., 2022) firstly uses ResNet-50 to recognize salient objects via a multi-class classification task, and then fuses them with question features via the BERT to get the answers. Self-paced curriculum learning (SPCL) (Yuan et al., 2022a) model learns VQA tasks in an easy-to-hard way, using simple questions to guide the learning of difficult questions. Spatial hierarchical reasoning network (SHRNet) (Zhang et al., 2023b) designs a hash-based spatial multi-scale module to adaptively capture key regions guided by question features. For complex relational reasoning, SOBA (Wang et al., 2024) leverages semantic features to guide the downstream VQA task. However, the semantic segmentation and VQA tasks are individually optimized, and directly training SOBA via multi-task losses inevitably causes mutual exclusivity in gradients. To address this, we propose EarthVQANet to explore simultaneous learning for segmentation and VQA, achieving mutual promotion.

3. EarthVQANet

An overview of EarthVQANet is shown in Fig. 2, which depicts: (a) hierarchical pyramid segmentation network, (b) semantic-guided attention, and (c) adaptive numerical optimization. The hierarchical pyramid segmentation network is designed to obtain land-cover semantic mapping results while also generating visual features and semantic guidance for the VQA task. The semantic-guided attention performs interactions between visual and language features, hierarchically fusing them for the final answer. The segmentation network and semantic-guided attention share a siamese encoder, contributing to universal presentation learning. The adaptive numerical optimization individually models the regression and classification tasks in VQA, introducing numerical sensitivity and sample balancing. Based on EarthVQANet, the segmentation and VQA tasks can be achieved under a universal architecture.

3.1. Hierarchical pyramid segmentation network

Motivation. The early study (Antol et al., 2015; Yang et al., 2016) directly predicted the final answers based on the raw images. They utilize CNN as the global feature extractor, and the pooling and flattening functions lose the spatial details. This limits the model's ability

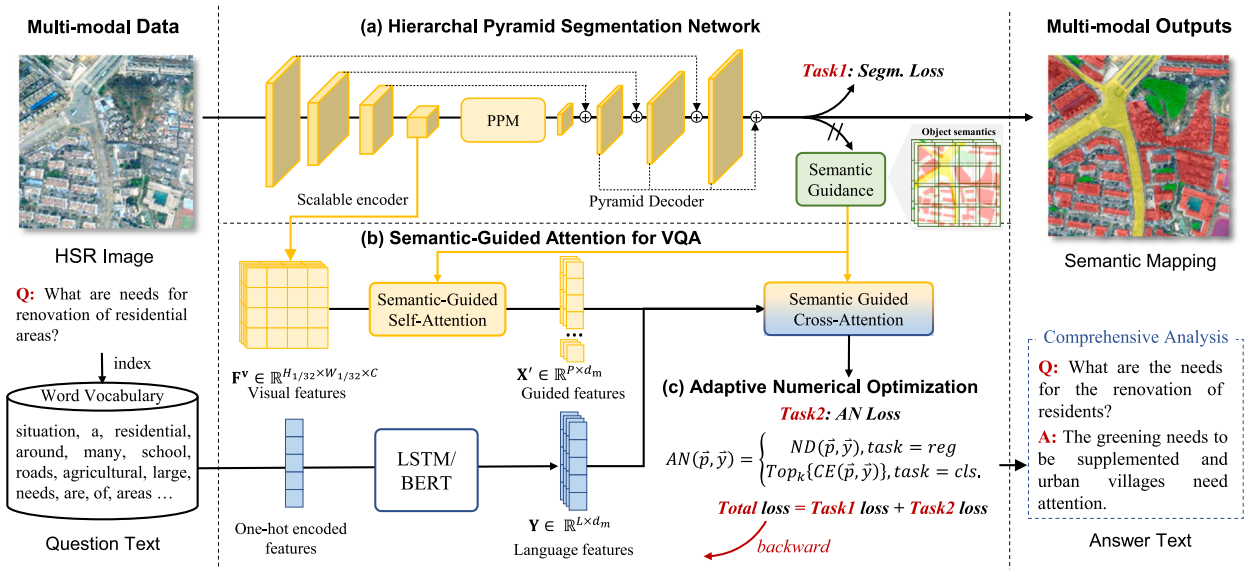


Fig. 2. Overview of the proposed universal segmentation and VQA framework (EarthVQANet). EarthVQANet mainly contains: (a) hierarchical pyramid segmentation network, (b) semantic-guided attention, and (c) adaptive numerical optimization.

to fit complex scenes with multi objects. To address this, BUTD (Anderson et al., 2018) introduced object features by using a detector (Faster-RCNN). The bounding boxes refer to the recognized objects, and local features within objects are aggregated with average poolings. These object features explicitly provide initial semantic representation, i.e., semantic categories, number of objects, etc. Inspired by BUTD, we aim to incorporate more spatial and refined features into the object features, thus enhancing the model's ability to represent complex scenes.

Design. To this end, we utilize a semantic segmenter (FCN) to generate more refined details. The segmenter is implemented with a hierarchical pyramid segmentation network, which compromises a siamese scalable encoder, a pyramid pooling module (PPM), and a pyramid decoder. We have tested the compatibility of the encoder with various advanced CNN and Transformer architectures, which allows for increased flexibility and accuracy in our analysis. The siamese encoder processes the input image hierarchically into four different spatial shapes, namely $1/4, 1/8, 1/16, 1/32$ with respect to the raw image. PPM (Zhao et al., 2017) is utilized to capture the multi-scale features from the encoder outputs. In the implementation, the encoder outputs are firstly processed by four average poolings with different scales (1, 2, 3, 6). The resulting multi-scale features are then reduced by 1×1 convolutions, resized to the same size, concatenated, and finally passed through a 3×3 convolution. Similar to the feature pyramid network (Kirillov et al., 2019; Ma et al., 2022), the pyramid decoder gradually restores the spatial resolution through a top-down pathway and skip-connections. The multi-scale features from different layers are then fused using a mean function, yielding the final semantic segmentation output.

The PPM and pyramid decoder capture the multi-scale features from different structure levels. Possessed with scalability and multi-scale processing ability, the hierarchical pyramid segmentation network provides a solid foundation of feature representation. This contributes significantly to the segmentation and VQA tasks.

3.2. Semantic-guided attention for VQA

Motivation. To leverage the object features provided by the segmentation network, we have designed a semantic-guided attention. As the segmentation features contain rich semantic and spatial details, the internal features of the encoder are shared. Besides, the segmentation outputs are utilized as semantic guidance due to the object semantic boundaries.

Design. The semantic-guided attention includes semantic-guided self-attention and semantic-guided attention. As is shown in Fig. 2, the semantic-guided self-attention is designed to enhance the visual features. The semantic-guided cross-attention aims to promote multi-modal interaction between visual and language features. The visual features $\mathbf{F}^v \in \mathbb{R}^{H_{1/32} \times W_{1/32} \times C}$ are generated from the outputs of scalable encoder, which preserves the spatial details and locations. H and W denote the spatial size of the raw image. C represents the feature dim. The VQA and segmentation shared the siamese encoder to learn universal representation and boost each other.

3.2.1. Semantic guidance

In order to introduce object semantic features, we utilize the segmentation output from the pyramid decoder to guide the VQA tasks. As is shown in Fig. 3, the generation process of semantic-guided involves (a) spatial scaling, (b) spatial dynamic weighting, and (c) spatial reduction. Because the segmentation output is only used for guidance, we interrupt the gradient flow to avoid the interference of the VQA task on the segmentation decoder. This guarantees that task-specific features are embedded in the pyramid decoder and semantic-guided attention separately. The related experiments are also analyzed in Table 5.

The spatial scaling aims to align the size of decoder output with the size of $\mathbf{F}^v$. Specifically, the scaling process is implemented with three 3×3 convolutions with strides of 2. Each convolution is followed with a batch normalization and a ReLU activation. The size of output is successively reduced, yielding $\mathbf{S}^v$. The spatial dynamic weighting is designed to modulate semantic features based on spatial characteristics. Inspired by the existing spatial attention mechanism (Woo et al., 2018), we firstly reduce the channel dimension from two aspects, i.e., max pooling and average pooling. The 7×7 convolution is then adopted to aggregate the spatial characteristics via a large receptive field. After aggregation, the key regions are enhanced and normalized via a sigmoid function. The spatial scores are used to weight $\mathbf{S}^v$ and finally flattened into $\mathbf{G}^v \in \mathbb{R}^{P \times d_m}$. $P = H_{1/32} \times W_{1/32}$ denotes the number of visual tokens and d_m is the hidden size.

3.2.2. Semantic-guided self-attention

To capture global relations between geospatial objects, we utilize vision transformer (Dosovitskiy et al., 2020) to construct self-attention module. As is shown in Fig. 4, the self-attention module consists of

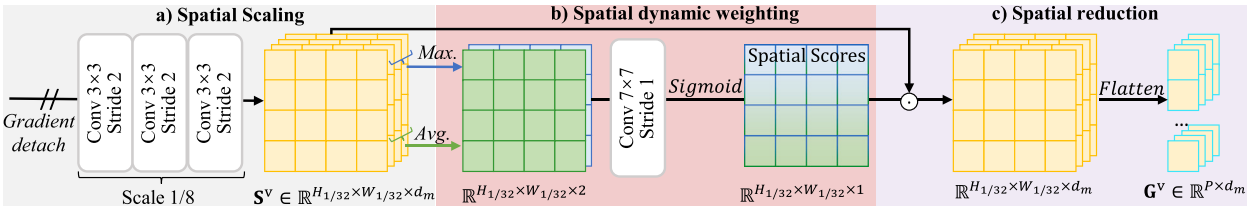


Fig. 3. Generation process of semantic guidance. The semantic guidance includes object semantic features which contributes the object consistency, spatial locations, etc. The process includes: (a) spatial scaling, (b) spatial dynamic weighting, and (c) spatial reduction.

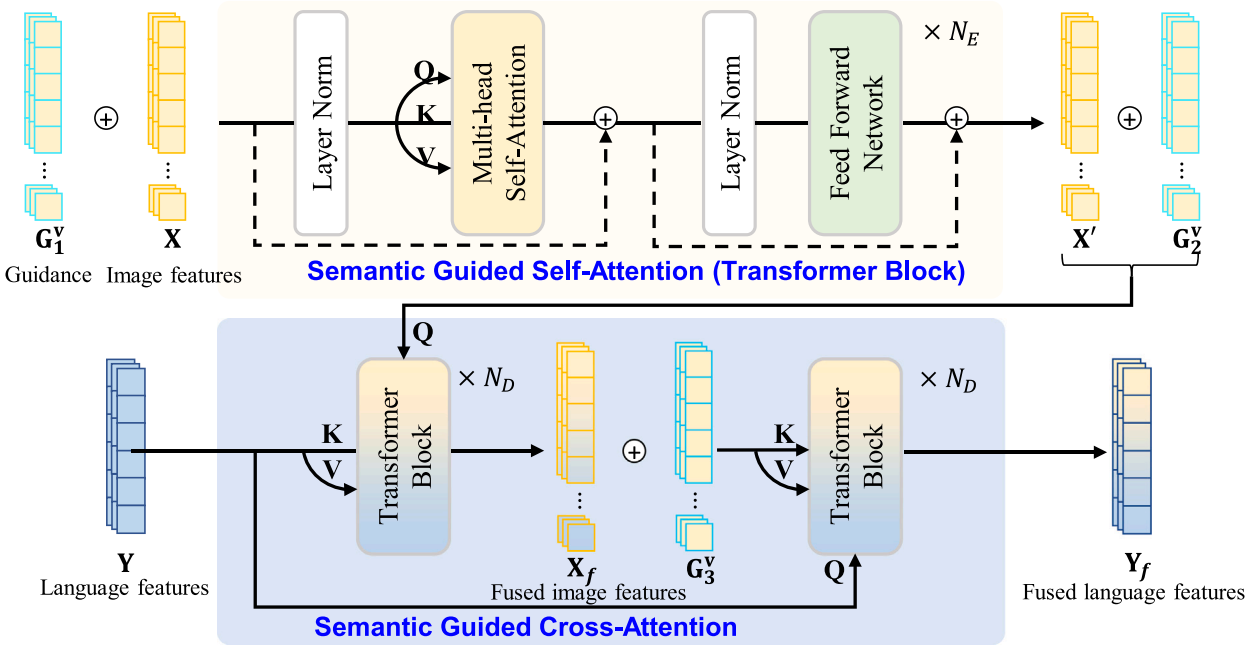


Fig. 4. Implementation of semantic-guided self-attention and semantic-guided cross-attention. The semantic guidance is integrated in different attention stages.

N_e Transformer blocks. Each Transformer block contains a multi-head self-attention (MSA) and a feed forward network (FFN), embedded with layer normalizations and residual connections. MSA utilizes M parallel heads with independent learnable weights. Different heads encode the feature similarities from diverse aspects. FFN consists of two linear transformation layers with a GELU activation, further performing hierarchical representation.

Before transformation, the visual features $\mathbf{X}$ were first summed up with the semantic guidance $\mathbf{G}^v$. At each Transformer block, the guided visual features are transformed into query, key, and value: $\mathbf{Q} = (\mathbf{X} + \mathbf{G}^v)\mathbf{W}^q, \mathbf{K} = (\mathbf{X} + \mathbf{G}^v)\mathbf{W}^k, \mathbf{V} = (\mathbf{X} + \mathbf{G}^v)\mathbf{W}^v$. $\mathbf{W}^q, \mathbf{W}^k, \mathbf{W}^v \in \mathbb{R}^{d_m \times d_v}$ are implemented with three linear projection layers and $d_v = d_m/M$ represents the feature dim of each head. The self-attention utilizes the feature relations between each patch to improve the representations of these patches. The process of self-attention formulates as follows:

$$Attention(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = softmax(\frac{(\mathbf{X} + \mathbf{G}^v)\mathbf{W}^q((\mathbf{X} + \mathbf{G}^v)\mathbf{W}^k)^T}{\sqrt{d_v}})(\mathbf{X} + \mathbf{G}^v)\mathbf{W}^v \tag{1}$$

There are multiple attention operations in parallel to jointly attend to different representation subspaces. After concatenation, a linear projection layer is utilized to fuse these outputs. Formally,

$$\begin{aligned} &MSA(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = Concat(h_1, \ldots, h_M)\mathbf{W}^O \\ &where \quad h_i = Attention(\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i) \end{aligned} \tag{2}$$

$\mathbf{W}^O \in \mathbb{R}^{Md_v \times d_m}$ represents the learnable weights of fusion layer. MSA dynamically models the relations of visual features between each object. FFN is designed to further improve the representations via non-linear transformations. Specifically, FFN contains two linear projection layers with a GELU in between. The process of FFN is formulated as follows:

$$FFN(\mathbf{X}) = GELU(\mathbf{X}\mathbf{W}_1)\mathbf{W}_2 \tag{3}$$

where $\mathbf{W}_1 \in \mathbb{R}^{d_m \times d_f}$ and $\mathbf{W}_2 \in \mathbb{R}^{d_f \times d_m}$ are projection weights. d_f represents the hidden dimension.

3.2.3. Semantic-guided cross-attention

The semantic-guided cross-attention is designed to fuse the visual and language features. The language features can be encoded with advanced methods, i.e., LSTM (Hochreiter and Schmidhuber, 1997), BERT (Kenton and Toutanova, 2019), etc. As for language encoding, a two-layer LSTM with a hidden size of 384, BERT-Base[1] and DistillBERT-Base[2] were adopted for comparison. The LSTM was randomly initialized and two BERT models were pre-trained on BooksCorpus (800M words) and English Wikipedia (2500M words). All the input language features were reduced to 384 via a linear projection. The visual clues are searched according to keywords in the question, and then summarized to the final answer. The cross-attention consists of two series of transformer blocks, where the fusion mechanisms are bidirectional. Similarly, the visual features from the self-attention are firstly fused with $\mathbf{G}_2^v$ to achieve semantic guidance. According to Fig. 4, the guided

[1] https://huggingface.co/bert-base-uncased.
[2] https://huggingface.co/distilbert-base-uncased.

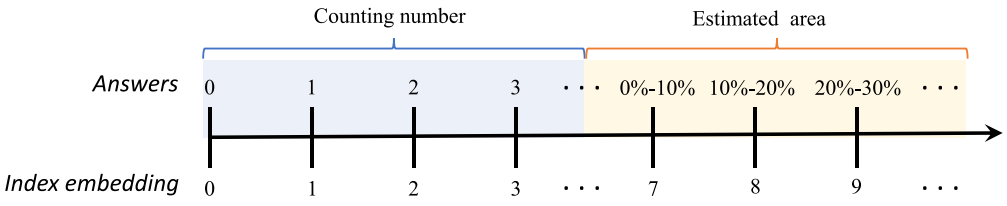


Fig. 5. The index embedding of counting answers. $\mathbf{y}$ is obtained from the answer's index.

features are transformed into a query vector $\mathbf{Q} = (\mathbf{X}' + \mathbf{G}_2^v)\mathbf{W}^q$, while key and value are generated from the language features $\mathbf{Y}$. This form of attention aggregates language features to enhance the visual representations $\mathbf{X}_f$. In the second stage, we regard language features as query, and visual features as key and value. The semantic guidance is once again performed, with $\mathbf{K} = (\mathbf{X}_f + \mathbf{G}_3^v)\mathbf{W}^k$, $\mathbf{V} = (\mathbf{X}_f + \mathbf{G}_3^v)\mathbf{W}^v$. The second transformer blocks then hierarchically aggregate visual features into language representations $\mathbf{Y}_f$. The keywords of interested objects will automatically collect corresponding visual features. After multi-modal reasoning, we fuse $\mathbf{X}_f$ and $\mathbf{Y}_f$ to obtain the final answer.

3.3. Adaptive numerical optimization

Motivation. The VQA tasks involve different types of questions, including classification and regression (object counting) questions. The existing research simply treats it as a multi-classification task and utilizes the standard cross-entropy (CE) loss for optimization. However, this optimization target is not suitable for regression tasks, due to its lack of distance awareness. Additionally, classification questions often have imbalanced simple and hard samples (Wen et al., 2021b), where simple questions dominate the optimization. To this end, we propose adaptive numerical optimization to automatically balance task and sample differences.

Design. We first revisit the multi-classification CE loss:

$$CE(\vec{p}, \vec{y}) = -\vec{y} \odot log(\vec{p}) = \sum_{i=1}^{class} -y_i log(p_i) \tag{4}$$

where $\vec{y}$ and $\vec{p}$ denote the ground truth and predicted probabilities. $\vec{y}$ is the one-hot encoded vector. To add distance awareness into CE loss, we design the modulating factor $d = \alpha|\mathbf{y}_{diff}|^\gamma = \alpha|\mathbf{y}_{pr} - \mathbf{y}_{gt}|^\gamma$. $\mathbf{y}_{pr}, \mathbf{y}_{gt}$ denote the number of prediction and ground truth for regression tasks. As the counting answers are embedded, their indexes represent the $\mathbf{y}$ (Fig. 5). $\alpha \geq 0$ and $\gamma \geq 0$ are hyperparameters to controls the intensity of distance penalty. d determines the distance penalty $d \propto \mathbf{y}_{diff}$. The numerical difference loss for regression tasks can be formulated as follows:

$$\begin{aligned} ND(\vec{p}, \vec{y}) &= -(1+d)\vec{y} \odot log(\vec{p}) \\ &= -(1+\alpha|\mathbf{y}_{diff}|^\gamma)\vec{y} \odot log(\vec{p}) \\ &= -(1+\alpha|\mathbf{y}_{pr} - \mathbf{y}_{gt}|^\gamma) \sum_{i=1}^{class} y_i log(p_i) \end{aligned} \tag{5}$$

As for the classification tasks, we designed hard example mining strategy to adaptively select the effective samples. Online hard example mining (OHEM) only optimizes hard samples with low confidence, which reflects low probability in the true label position. Hence, OHEM filters the easy samples with a probability threshold. However, a fixed probability threshold can lead to training instability as the number of classified samples in mini-batches changes. To address this, we modify the filter method to select a fixed number of hard samples with high losses, controlled by parameter k. $k \in (0, 1]$ denotes the remained ratio for hard samples with top losses. This allows for the selection ratio of hard samples to be adjusted adaptively. For example, it is assumed that one mini-batch has 12 classification samples with their losses: {0.81, 0.23, 0.34, 0.54, 0.63, 0.72, 0.12, 0.33, 0.54, 0.44, 0.09, 0.14}. If $k = 0.8$, we only select $12 \times 0.8 = \lfloor 9.6 \rfloor = 9$ samples with high losses to optimize. In this case, the easy samples {0.09, 0.12, 0.14} are filtered out. As the training process continues, the loss of each sample will be different in each epoch. The model automatically selects the samples that are most worthy of training. The universal version of our adaptive numerical loss is formulated as follows:

$$AN(\vec{p}, \vec{y}) = \begin{cases} ND(\vec{p}, \vec{y}), & \texttt{task} = reg. \\ Top_k\{CE(\vec{p}, \vec{y})\}, & \texttt{task} = cls. \end{cases} \tag{6}$$

The proposed adaptive numerical optimization combines the classification and regression VQA tasks in one unified loss. This automatically balances task and sample differences during the training without any additional parameters.

The simplified version of the proposed multi-task framework is shown in Fig. 6. By sharing the encoder weights, the segmentation features support the VQA tasks according to the forward path. Simultaneously, the VQA loss supervises the encoder optimization via the backward path, which then refines the segmentation results. By implicitly modeling constraints, 'There are buildings exist in the scene' will increase the probability of buildings appearing in segmentation, and 'The area of buildings occupies 10%–20%' controls the number of pixels that are predicted to 'building'.

4. EarthVQA dataset

The EarthVQA dataset (Wang et al., 2024) was extended from the LoveDA dataset (Wang et al., 2022b), which was originally designed for the land-cover segmentation task. The images were collected from Wuhan, Nanjing, and Changzhou cities including 18 urban and rural administrative districts (536.15 km^2). With the addition of matched QA pairs based on the city planning requirements, the EarthVQA dataset contains 6000 HSR images, 6000 semantic land-cover masks, and 205,593 QA pairs. The size of each image is 1024 × 1024, with a spatial resolution of 0.3 m.

4.1. Dataset statistics

Fig. 7 shows the distributions of questions and each question owns a considerable number of samples. Each urban image is equipped with 42 QA pairs, and each rural image has 29 QA pairs. Because the urban (rural) sample has same questions and similar answers, we evenly divide regions into train (4 urban, 4 rural), val (2 urban, 2 rural), test (3 urban, 3 rural) sets to ensure the balance of QAs, following dataset standards. The whole dataset was divided into *train* (2522 images with 88 166 QA pairs), *val* (1669 images with 57 202 QA pairs) and *test* (1809 images with 62 744 QA pairs) sets. The EarthVQA dataset includes six types of questions, i.e., basic judging, basic counting, relational-based judging, relational-based counting, object situation analysis, and comprehensive analysis. The semantic categories include background, building, road, water, barren, forest, agricultural and playground.

Fig. 8 shows more detailed distributions of questions and answers. The number of unique answers in the EarthVQA dataset is 166. The distributions of questions with different types are shown in Fig. 8(a). The relational-based judging has the most samples (83.2k) and the relational-based counting occupies the minimal samples (6k). This is because relational-based judging has many concerned questions (Fig. 7). Each question type includes a different number of questions,

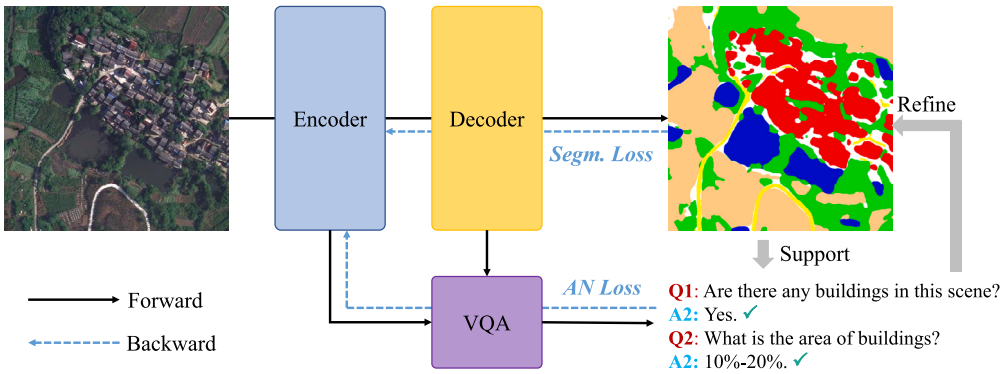


Fig. 6. The simplified version of our multi-task framework. By sharing the encoder weights, the segmentation features support the VQA tasks according to the forward path. Simultaneously, the VQA loss supervises the encoder optimization via the backward path, which then refines the segmentation results.

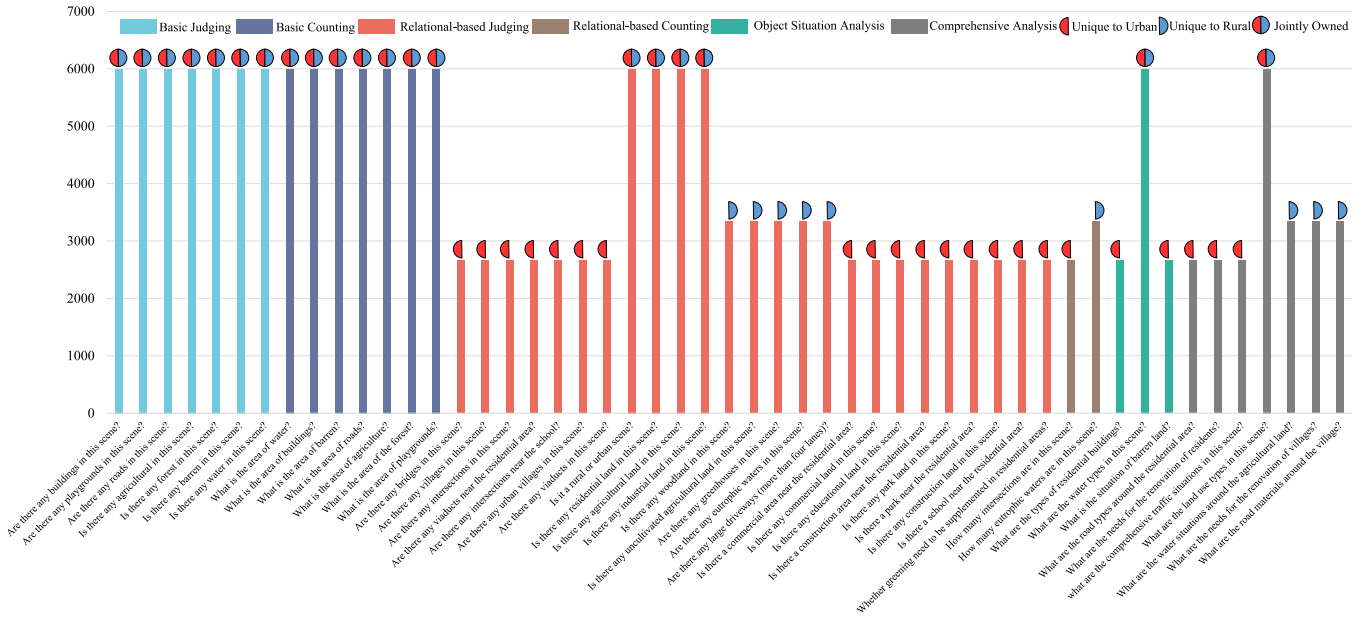


Fig. 7. The distributions of questions in the EarthVQA dataset (Wang et al., 2024). Based on city planning requirements, the EarthVQA dataset includes six types of questions, i.e., basic judging, basic counting, relational-based judging, relational-based counting, object situation analysis, and comprehensive analysis. The distributions of questions are relatively balanced, ensuring comprehensive and adequate training and evaluation.

further intensifying the differentiation in difficulty of question types. The answer distributions are complex and reveal many practical challenges for the HSR VQA task. As shown in Fig. 8(b), the counting answers showcase a long-tail distribution. As for judging questions, the answers in (d) and (e) show different distributions. The 'Yes' occupies a larger proportion in basic judging but 'No' has more samples in relational-based judging. As for object analysis situation and comprehensive analysis, some questions such as (c) and (f) have relatively balanced answer distributions but (g) have imbalanced answer distributions. In conclusion, each question has different distributions of answers, which brings more challenges when faced with the actual Earth's environment. These challenges inspire some new directions to advance complex reasoning in remote sensing VQA. As for imbalanced answer distributions, it is worth using the denoising diffusion probabilistic model (Ho et al., 2020) to generate samples with rare answers. As for fine-grained answers in object situation analysis, contrastive learning (Wang et al., 2021b) can be explored to enhance the model's discriminative ability.

Fig. 9 shows two samples from urban and rural scenes, respectively. The urban scenes focus on the residents, traffic situation, green spaces, water sources, urban villages, etc. The rural scenes focus on water governance, road improvement, agricultural cultivation, etc.

4.2. Annotation procedure

The basic questions are automatically generated from semantic masks, similar to RSVQA (Lobry et al., 2020) and others are manually annotated. Fig. 10 shows some annotation procedures of representative samples. **(1) Basic judging**. The basic judging questions focus on the presence or absence of land cover types. Each semantic category is separately queried, and the answer is generated from the corresponding mask. **(2) Basic counting**. The basic counting questions are used to estimate the area of features. Considering that each image has the same spatial resolution of 0.3 m, we evenly divided it into ten intervals. The upper bound of area is $1024 \times 1024 \times 0.3$ (m) $\times$ 0.3 (m) $=$ 94371.84 m^2 **(3) Object situation analysis**. As for 'What are the types of residential buildings?', the candidate's answers include commercial buildings, private buildings and no residential buildings. The commercial buildings denote houses contracted and sold by real estate developers, which often show consistent appearances and neat layouts. The private buildings are often built by individuals and have inconsistent appearances as well as heights. It is possible that there are both commercial housing and private housing in the same scene. **(4) Comprehensive analysis**. As for 'What are the roads around the village?', we first search for the village, which is formed of compact

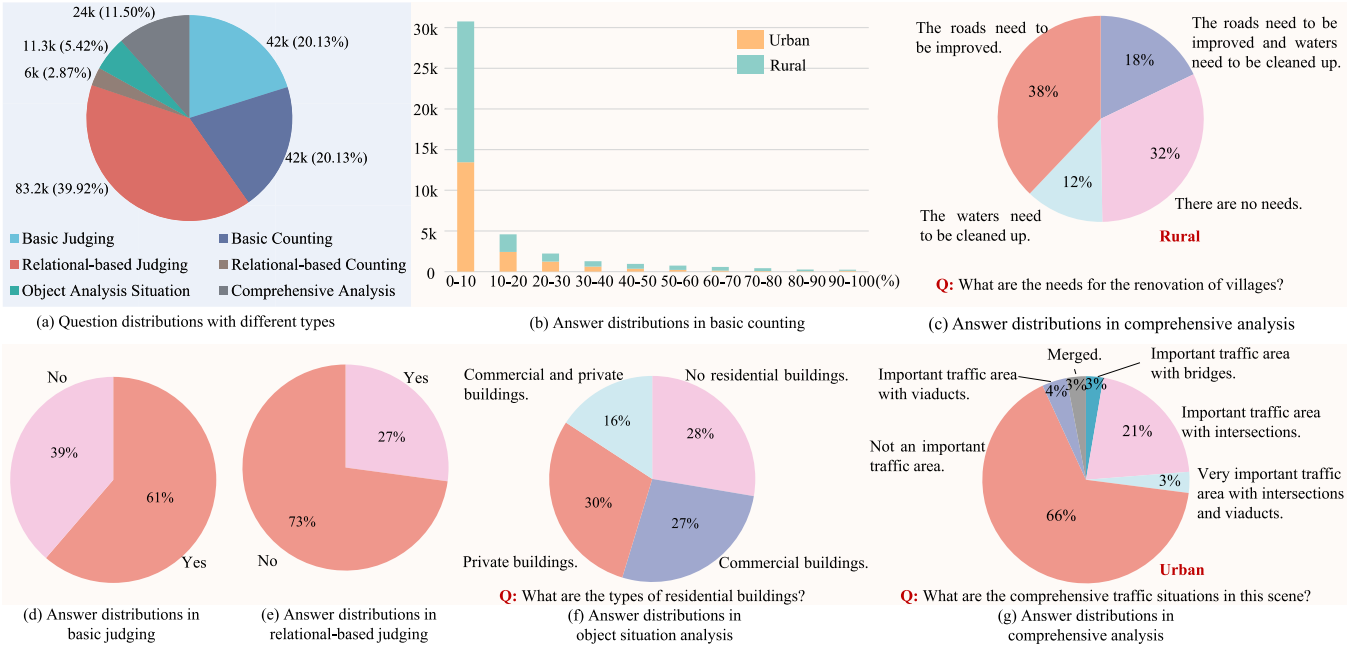


Fig. 8. More detailed statistical distributions of questions and answers.

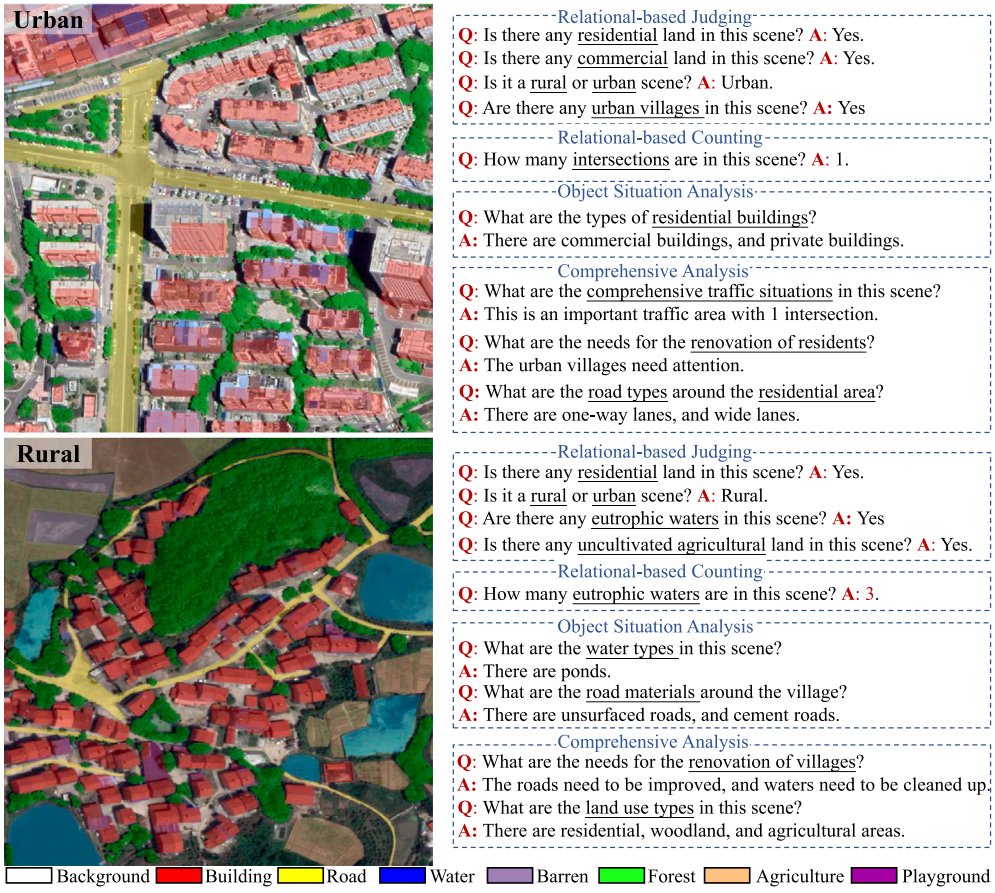


Fig. 9. Representative samples from the EarthVQA dataset. The urban and rural scenes focus on the different schemes. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

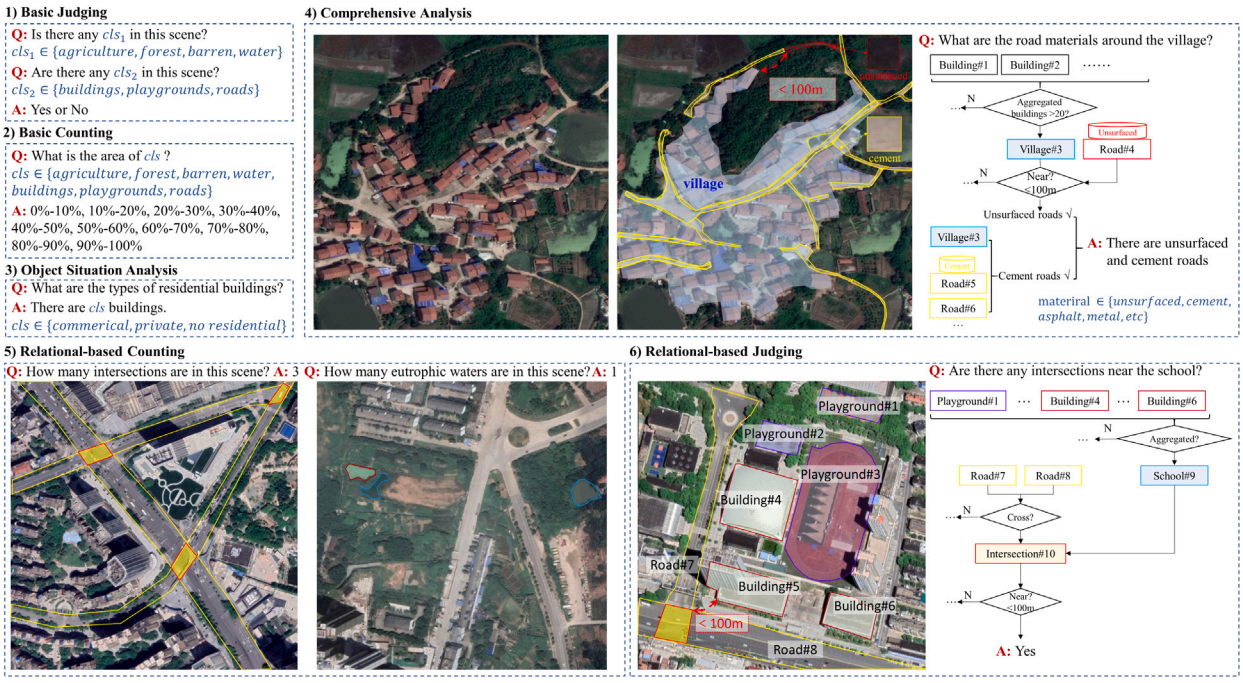


Fig. 10. Representative samples from the EarthVQA dataset. The urban and rural scenes focus on the different schemes.

buildings (more than 20 buildings). The aggregated buildings form a polygon of the village. Most of roads are cement but a small section of road has not yet been paved. These roads are near (<100 m) to the village, thus we obtain the final answer 'There are unsurfaced and cement roads'. **(5) Relational-based counting**. The relational-based counting requires reasoning spatial or semantic relations that are not included in original semantic masks. For 'How many intersections are in this scene?', the annotators need to judge the topologies of roads and then count the number of crossed parts. For 'How many eutrophic waters are in this scene?', the annotators need to judge the sub-properties of all waters and count the number of waters with abnormal spectra. **(6) Relational-based judging**. The relational-based judging involves spatial or semantic reasoning with several objects. As illustrated in 'Are there intersections near the school?', a stadium, several teaching buildings, and playgrounds are close to each other, forming the school. Besides, two roads are crossed to form an intersection. The annotators used the ArcGIS toolbox to calculate the polygon-to-polygon distance between the school and the intersection, obtaining 23 m < 100 m. Hence, the final answer is 'Yes'. Besides, some land-use types (commercial, industrial, educational) are determined using OSM data as an auxiliary. The distance judgment threshold is unified to 100 m and our dataset does not involve ambiguous questions such as geographical orientations.

The annotation thresholds can be summarized as follows: (1) The distance threshold judging 'near' is 100 m; (2) 'more than 20 compact buildings' form a resident; (3) The private buildings in residents have inconsistent appearances as well as heights; (4) The commercial buildings show consistent appearances and neat layouts. (5) The waters with green algae and other floating vegetation are eutrophic waters. (6) Some land-use types with socioeconomic attributes (commercial, industrial, etc.) are determined using OSM properties as auxiliary. (7) If the leaf area index (vegetation area/total area) in the residential area is less than 30%, it needs to be supplemented.

5. Experiments

5.1. Experimental settings

Evaluation metrics. Since our framework includes both semantic segmentation and VQA tasks, we report their performances separately using different metrics. For semantic segmentation tasks, we report the overall performance using mean Union and Intersection (mIoU) as recommended in Wang et al. (2022b). For VQA tasks, we report classification accuracy and root-mean-square error (RMSE) as commonly used (Lobry et al., 2020). The overall performances include the overall accuracies (OA) and overall root-mean-square error (OR).

Reference methods. For the fairness nine general VQA methods and two remote sensing VQA methods are selected for reference. The general VQA methods include SAN (Yang et al., 2016), MAC (Hudson and Manning, 2018), BUTD (Anderson et al., 2018), BAN (Kim et al., 2018a), MCAN (Yu et al., 2019), LXMERT (Tan and Bansal, 2019), D-VQA (Wen et al., 2021b), BLIP-2 (Li et al., 2023a), Instruct-BLIP (Dai et al., 2024). The remote sensing methods are RSVQA (Lobry et al., 2020), RSIVQA (Zheng et al., 2021) and SOBA (Wang et al., 2024). Following the original settings, BLIP-2 and Instruct-BLIP used the pre-trained ViT-g/14 and FlanT5$_{XL}$ as encoders. The Q-Former was trained to bridge the multi-modal features for conditional generation. As BUTD, BAN, D-VQA, MCAN, and LXMERT require local semantic features as inputs, we adopt the proposed multi-task learning strategy fairly, and the visual features are generated by the hierarchical pyramid segmentation network. The CNN encoders are adopted as ConvNeXt-Tiny (Liu et al., 2022) as default, and other settings are identical to their original literature.

Training implementations. As for the EarthVQA dataset, all VQA methods are trained for 55k steps with a batch size of 16. For multi-task learning, we only trained the segmentation network in the first 15k steps for initialization, and the rest is utilized for joint training. The data augmentations include random flipping, rotation, and color jittering. Consistent with LoveDA dataset (Wang et al., 2021a), we set scales = {0.5, 0.75, 1.0, 1.25, 1.5, 1.75} for multi-scale training. The 512 × 512 patches are randomly cropped as segmentation inputs and all images are resized into 768 × 768 for VQA inputs for information integrity. We set the initial learning rate to 1e-4 and use a 'poly' schedule for adaptation. To analyze the contributions of each proposed module, we have conducted comprehensive analysis experiments as follows. Because the urban and rural scenes have different answer distributions according to their characteristics, we evenly divided the urban and rural scenes into Train/Val/Test sets (Wang et al., 2021a) to ensure the diversity of model training and evaluation.

Table 2
Compared results with other VQA methods on EarthVQA^{test} set (Kim et al., 2018b).

Method	↑mIoU (%)	↑Accuracy (%)						↑OA (%)	↓RMSE		↓OR
		Bas Ju	Rel Ju	Bas Co	Rel Co	Obj An	Com An		Bas Co	Rel Co	
Only Segm.	57.27	–	–	–	–	–	–	–	–	–	–
• *General methods*											
SAN (Yang et al., 2016)	–	88.73	80.61	77.16	64.98	52.73	42.70	75.52	1.0253	1.2197	1.0516
MAC (Hudson and Manning, 2018)	–	82.71	78.60	71.82	55.64	45.71	38.10	71.19	1.4250	1.3576	1.4167
BUTD (Anderson et al., 2018)	57.11	90.36	82.00	79.45	67.58	55.89	46.80	77.55	0.7919	1.1787	0.8499
BAN (Kim et al., 2018b)	56.46	90.83	82.27	78.66	63.93	48.65	45.14	76.74	0.9155	1.2367	0.9615
D-VQA (Wen et al., 2021b)	56.95	90.07	81.94	79.47	66.26	58.09	45.88	77.54	0.8233	1.2071	0.8805
MCAN (Yu et al., 2019)	57.07	**90.62**	81.03	79.90	68.14	58.06	47.09	77.53	0.7977	1.1208	0.8449
LXMERT (Tan and Bansal, 2019)	56.35	90.00	81.93	80.54	**68.26**	54.87	45.07	77.46	0.7980	1.1731	0.8539
BLIP-2 (Li et al., 2023a)	–	88.13	81.92	70.26	58.58	42.72	28.34	71.07	1.8790	1.3200	1.8186
Instruct-BLIP (Dai et al., 2024)	–	89.67	79.69	76.96	63.34	59.72	45.68	75.25	0.7994	1.2170	0.8627
• *RS methods*											
RSVQA (Lobry et al., 2020)	–	82.38	79.14	71.80	55.42	42.98	36.27	70.95	1.5162	1.3605	1.4976
RSIVQA (Zheng et al., 2021)	–	87.83	80.66	77.03	64.71	54.02	42.35	75.41	0.9715	1.2206	1.0060
SOBA[a] (Wang et al., 2024)	57.17	91.10	81.56	80.84	67.09	58.34	47.94	78.24	0.7642	1.1258	0.8182
SOBA-MT (Wang et al., 2024)	56.01	89.82	81.18	79.69	**68.69**	58.34	47.07	77.42	0.7790	1.1021	0.8263
EarthVQANet (w/o Segm.)	–	89.23	80.38	77.42	65.00	53.67	44.82	76.47	0.9934	1.2045	1.0214
EarthVQANet	**57.43**	90.52	**82.09**	**81.90**	67.92	**59.06**	**48.64**	**78.54**	**0.7490**	**1.0808**	**0.7980**

The abbreviations are: Bas Ju (Basic Judging), Rel Ju (Relational-based Judging), Bas Co (Basic Counting), Rel Co (Relational-based Counting), Obj An (Object Situation Analysis), Com An (Comprehensive Analysis). OA (Overall Accuracy), OR (Overall RMSE).
[a] Denotes the cascade training for SOBA in the original setting.

5.2. Comparative experiments

The comparative multi-task results on the EarthVQA dataset are presented in Table 2. As the segmentation performance provides a basis for VQA, the Only Segm. setting shows that the proposed hierarchical pyramid segmentation network obtains a baseline mIoU of 57.27% on the segmentation task. The methods that directly utilize global visual features, i.e., SAN, MAC, RSVQA, and RSIVQA, achieve low VQA performances on both classification and regression tasks. With the global features as inputs, it is hard for the EarthVQANet (w/o Segm.) to answer the relational-based questions. As for Instruct-BLIP and BLIP-2, the visual encoders pre-trained on ImageNet are not suitable for remote sensing VQA tasks. As for Basic counting task, our proposed EarthVQANet has achieved the best accuracy (81.90%) and RMSE (0.7490). As for relational-based counting task, our proposed EarthVQANet (Acc = 67.92%, RMSE = 1.0808) fails to exceed BAN (Acc = 69.52%, RMSE = 1.0656). This may because the relational-based counting samples in our dataset are limited (only occupies 2.87% according to Fig. 8(a)) so our numerical sensitive AD loss may cause some overfit effects. In the future, we will develop some regular constraints to improve AD loss and reduce its overfitting risk. Implemented by the proposed multi-task learning strategy, the existing Faster-RCNN-based methods, i.e., BUTD, BAN, D-VQA, and MCAN, are also compatible with our proposed visual features, significantly exceeding the global methods. This demonstrates the necessity and importance of high-quality visual features for complex VQA tasks. Our proposed EarthVQANet generally outperforms the existing general and remote sensing methods. Specifically, for object counting tasks, compared with the best referenced method (MCAN, OR = 0.8449), EarthVQANet significantly reduces the error by 5%. Directly training SOBA simultaneously (SOBA-MT) lost lots of performance on segmentation and VQA tasks, because directly sharing image encoder and decoder gradients causes the optimization contradiction (Vandenhende et al., 2022). Hence, we firstly adopts PPM for no-linear projections to weaken the negative gradient interactions. The semantic-guided attention leverages residual connections to further guarantee the original VQA gradient flow. Moreover, the gradients from semantic guidance are blocked to avoid the effects of the global VQA semantics for image decoder training. In conclusion, EarthVQANet also improves the segmentation accuracy, proving its potential to achieve the mutual promotion of segmentation and VQA. The universal and task-specific semantics are effectively encoded in our siamese encoder and different decoders. With the roaring of large multi-modal models, exploring the application in remote sensing VQA is very promising. There exist some new directions to design remote sensing-specific large-modal models: (1) Visual features alignment. As the pre-trained vision encoders are pre-trained using CLIP (Radford et al., 2021) or MAE (He et al., 2022) based on natural images, it is necessary to align the general visual features with remote sensing images. (2) Task-driven application. The current large language models (LLMs) are constructed for general scenes via casual language modeling (Liu et al., 2024) or instruction tuning (Dai et al., 2024). It is worth exploring practical requirements and scenes in remote sensing tasks (such as city planning, disaster assessment, etc.)

In order to show the comparative results intuitively, two representative samples are selected from urban and rural scenes for visualizations (see Fig. 11). As for the urban scene, all referenced methods fail to recognize the intersection in the judging question (#**Q1**) but part of them succeed in answering the counting question (#**Q2**). The roads in the north-south direction are blocked by trees and the school scene is not obvious, which increases the difficulty of spatial reasoning. Compared with the referenced methods, EarthVQANet successfully achieves the right answers to these two questions and has good reasoning consistency. #**Q3** requires recognition of high-level functional zones, and the global fusing methods (RSVQA and SAN) generate wrong answers due to the lack of object details. MCAN and BUTD wrongly decided that there is a viaduct (#**Q4**) due to shadows cast by tall buildings, but EarthVQANet successfully avoids this inference error. The selected rural scene represents relatively simple distributions, the methods (MCAN, BUTD and EarthVQANet) with local visual features also outperforms than global fusion methods (RSVQA and SAN). The proposed EarthVQANet performs well in both semantic extraction and knowledge reasoning stages and exhibits high reasoning consistency across different questions.

5.3. Segmentation network analysis

Scalable image and language encoders. As the image encoder determines the universal representation ability, we evaluate its scalability with varied CNN and Transformer architectures, including ResNet (He et al., 2016), HRNet (Wang et al., 2020), ConvNeXt (Liu et al., 2022),

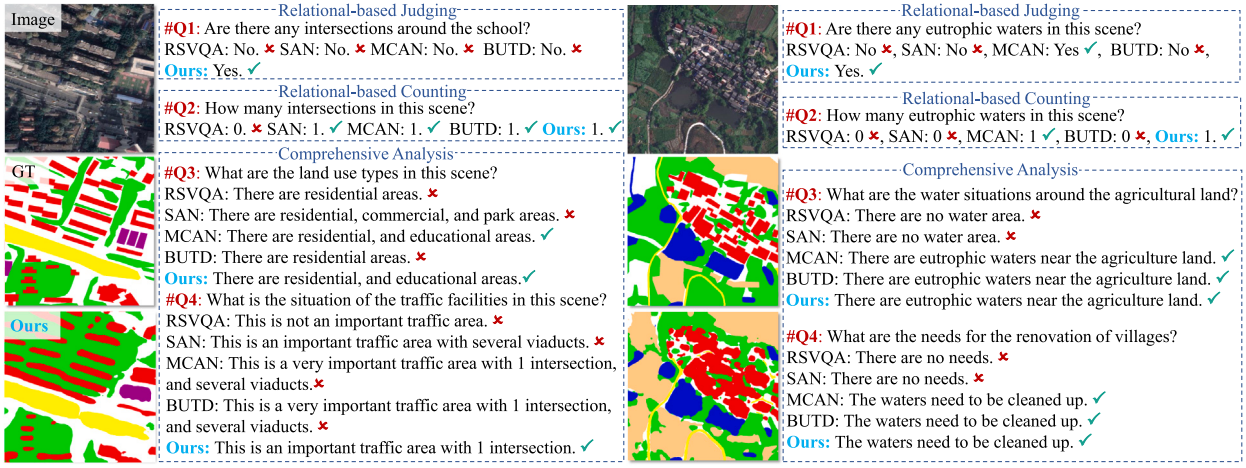


Fig. 11. Visualizations of predicted samples from urban and rural areas. Several typical questions are selected for comparison. EarthVQANet shows strong performances on questions of varying difficulty.

Table 3
Encoder ablation study.

Image	Language	Param(M)	↑mIoU (%)	↑OA (%)	↓OR
ResNet-50	LSTM	65.95	54.56	77.54	0.8276
ResNet-50[a]	LSTM	65.95	52.48	75.08	0.8674
ResNet-101	LSTM	80.06	55.12	77.76	0.8175
ResNet-101[a]	LSTM	80.06	53.02	75.78	0.8658
HRNet-W32	LSTM	55.41	55.67	77.89	0.8691
HRNet-W40	LSTM	71.88	55.72	78.08	0.8463
ConvNeXt-Tiny	LSTM	65.64	**57.43**	78.54	**0.7980**
ConvNeXt-Tiny	BERT-Base	173.23	57.28	**78.88**	0.8101
ConvNeXt-Tiny	DistillBERT-Base	130.11	57.27	78.75	0.7983
ConvNeXt-Small	LSTM	87.28	57.34	78.62	0.8040
ConvNeXt-Small	BERT-Base	194.87	57.14	78.84	0.8088
ConvNeXt-Small	DistillBERT-Base	151.75	57.25	78.82	0.8101
MiT-B2	LSTM	60.17	56.36	77.25	0.8389
MiT-B3	LSTM	74.72	56.83	77.59	0.8287
Swin-Tiny	LSTM	65.34	56.10	77.35	0.8481
Swin-Tiny	BERT-Base	172.93	54.82	77.80	0.8403
Swin-Tiny	DistillBERT-Base	129.81	55.13	77.78	0.8351
Swin-Small	LSTM	86.66	56.34	77.52	0.8498
Swin-Small	BERT-Base	194.25	56.02	78.03	0.8452
Swin-Small	DistillBERT-Base	151.13	56.10	78.05	0.8422

[a] Denotes the frozen ResNets of Faster-RCNN pre-trained on Visual Genome.

MiT (Xie et al., 2021), and Swin-Transformer (Liu et al., 2021). Table 3 presents that the advanced encoders achieve higher segmentation results and lead to better VQA performances. For similar structures, a larger number of parameters tend to result in higher VQA performance. Therefore, given sufficient computing power and time, larger image encoders are recommended. Compared with LSTM, BERT-Base further improves the VQA performances due to powerful Transformer structure and pre-trained weights on an external large corpus. As a student model of BERT, DistillBERT contains fewer parameters and achieves compatible performances. Besides, we also added comparative experiments with the general local feature extractors (Tan and Bansal, 2019; Anderson et al., 2018) pre-trained on Visual Genome (Krishna et al., 2017). The compared results in Table 3 demonstrate the effectiveness of the remote sensing visual feature extractors. Ablation experiments demonstrate the good scalability and compatibility of the proposed EarthVQANet.

Hierarchical pyramid decoder. The hierarchical pyramid decoder aims to capture multi-scale features and achieves accurate segmentation results and pseudo semantic guidance. The ablation study of two pyramid modules has been performed, with the results presented in Table 4. Compared with the naive decoder, the PPM significantly enhances segmentation results by 1.25% mIoU and VQA accuracies by 1.26% OA. The pyramid decoder (PD) structure boosts both the segmentation (+2.68% mIoU) and VQA (+0.71% OA) performances. Because PPM and PD enhance multi-scale features at the cell and structure layers respectively, their fusion brings complementary gains. Increasing the inner channels of the decoder slightly improves VQA performance, but it may also lead to overfitting. Overall, the combination of PPM and PD achieves effective information extraction and exhibits strong compatibility with VQA tasks.

Table 4
Ablation study of hierarchical pyramid decoder.

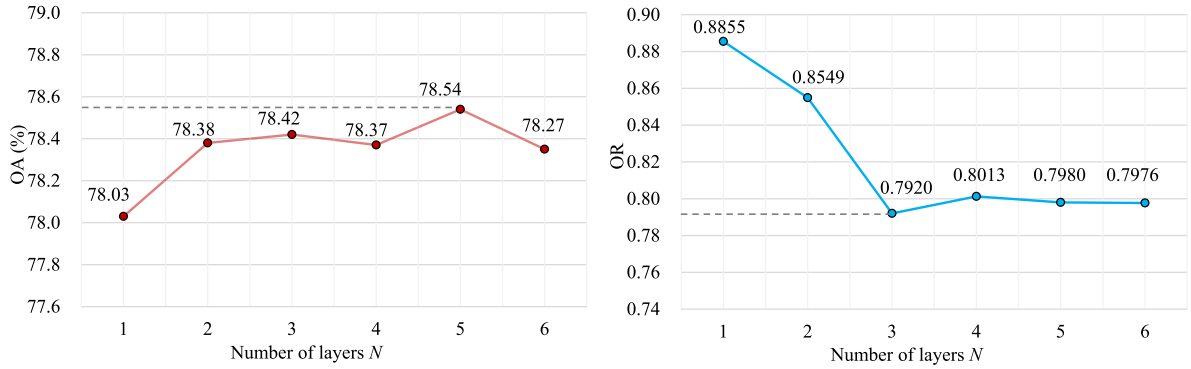
Structure	PPM	PD	#Channel	Params (M)	↑mIoU (%)	↑OA (%)	↓OR
Naive decoder			256	46.80	53.66	76.36	1.5761
+PPM	✓		256	61.35	54.91	77.62	0.8415
+PD		✓	256	51.34	56.35	78.26	0.8313
+PPM+ PD (Ours)	✓	✓	256	65.64	**57.43**	78.54	0.7980
			384	69.34	57.27	**78.71**	**0.7960**
			512	74.21	56.71	78.38	0.9313

Table 5
Ablation study of semantic guidance.

Semantic guidance	Types	Gradient	↑mIoU (%)	↑OA (%)	↓OR
SVL	–	×	56.67	77.55	0.8650
S_GVL	SG	×	57.17	78.15	0.8481
S_GVL	SG	✓	56.83	78.04	0.8673
SV_GL	CG	×	56.86	78.05	0.8410
SVL_G	CG	×	57.39	77.36	0.8636
SV_GL_G	CG	×	57.18	78.16	0.8290
SV_GL_G	CG	✓	56.79	77.32	0.8269
$S_GV_GL_G$	SG+CG	×	57.43	**78.54**	**0.7980**
$S_GV_GL_G$	SG+CG	✓	56.88	77.91	0.8304
$S_{G5}V_GL_G$	SG+CG	×	57.25	78.09	0.8066
$S_GV_{G5}L_G$	SG+CG	×	**57.44**	78.23	0.8215
$S_GV_GL_{G5}$	SG+CG	×	57.07	78.01	0.8241
$S_{G5}V_{G5}L_{G5}$	SG+CG	×	56.81	77.90	0.8448

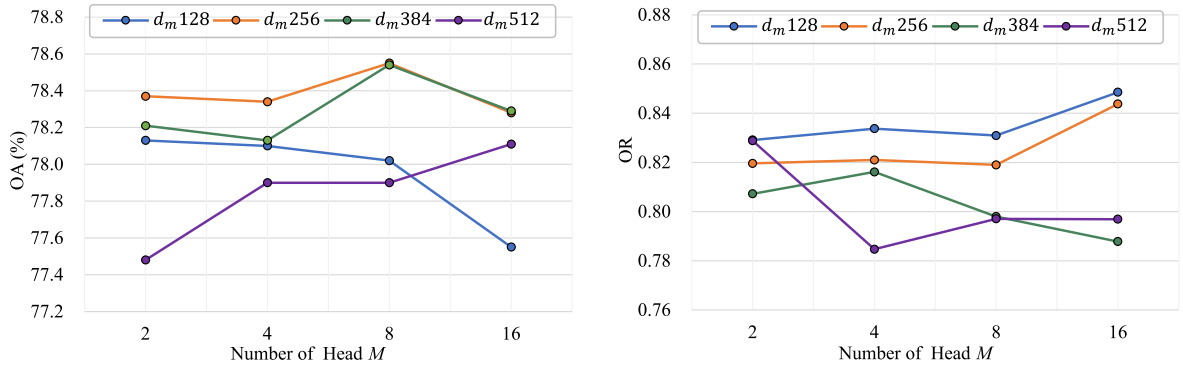
5.4. Semantic-guided attention

Semantic guidance. As semantic guidance introduces object semantics and spatial details, we have performed comparative experiments for guidance types and positions to evaluate the effects. Fig. 4 illustrates that the default setting comprises three types of guidance. The three guidances are located respectively before self-attention modules, one-stage cross-attention modules, and two-stage cross-attention



(a) OA with varied layers. (b) OR with varied layers.

Fig. 12. VQA overall and counting accuracies with varied number of attention layers. The performances firstly improve as the number of layer increase and gradually convergence.



(a) OA with varied number of heads and feature dim. (b) OR with varied number of heads and feature dim.

Fig. 13. VQA overall and counting accuracies with varied number of heads and feature dim. The performances firstly improve as the number of layer increase and gradually convergence.

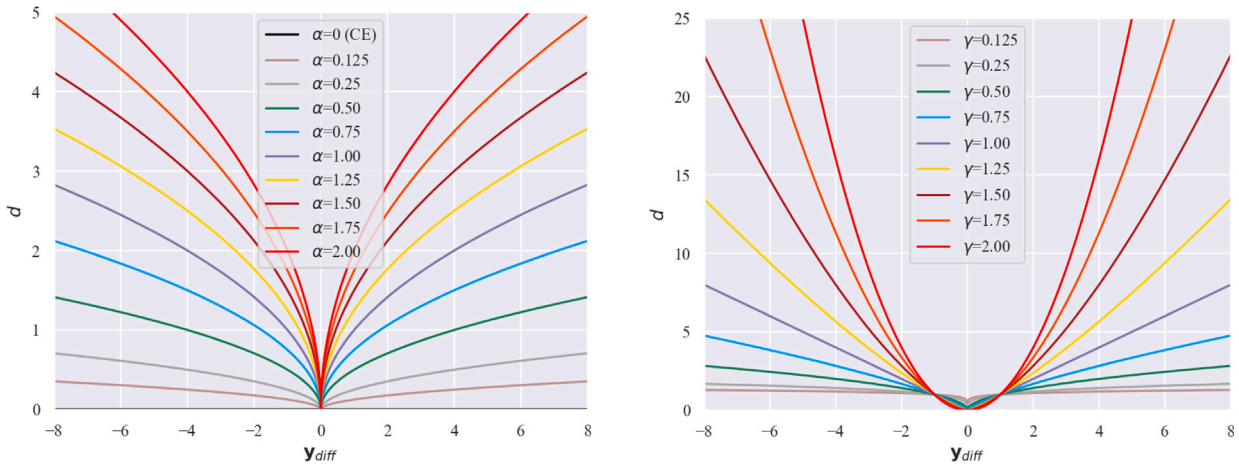
modules. This configuration can be expressed as $S_G V_G L_G$, where G represents the guidance, and V and L represent the bidirectional cross-attentions for visual features as query and language features as query, respectively. Since each attention module is composed of five transformer blocks, we also added guidance before each block to test its effects, denoted as $G5$. For example, V_G indicates that 'As for five cross-attention blocks which use visual features (V) as a query, only the first cross-attention adopts the semantic guidance (G)'. V_{G5} denotes that all five cross-attentions adopt the semantic guidance. Table 5 presents the results of the ablation study of semantic guidance. The addition of semantic guidance on self-attention ($S_G V L$) significantly improves the VQA overall accuracy (+0.61% OA). As for the bidirectional cross-attention, the guided visual features ($S V_G L$) also boost the VQA performance with visual features as query. Without pre-enhancement, the late guidance $S V L_G$ may cause semantic confusion and lead to negative effects on the VQA performance. The combination of different guidances ($S_G V_G L_G$) results in the best VQA performance at a system level. In any case, adding a gradient return to the semantic guidance causes negative effects. This is due to the direct supervision from VQA tasks conflicting with the segmentation representation. Besides, intensive guidance cannot lead to further improvements but induce overfitting. In conclusion, we selected $S_G V_G L_G$ configuration for the semantic guidance.

Ablation on number of attention layers. The semantic guidance attention includes self-attention and cross-attention, each consisting of several transformer blocks. Since the number of attention layers directly impacts the VQA representation capability, we varied it to evaluate the effects. For simplicity, we set $N = N_E = N_D$. Fig. 12 shows the VQA overall and counting performances with varying numbers of layers. As the number of layers increases, the performance initially improves and gradually converges. The best OA and OR are achieved when $N = 5$ and $N = 3$, reaching 78.54% and 0.792, respectively. When faced with a new dataset, we can gradually deepen the layers and find a good trade-off between accuracy and efficiency.

Ablation on number of heads and feature dim. As key hyperparameters for transformer blocks, the number of heads (M) and feature dim (d_m) determine the width of the network. To evaluate their effects, we conducted ablation experiments with varied $M = 2, 4, 8, 16$ and $d_m = 128, 256, 384, 512$. The comparative results in Fig. 13(a) show two different trends. When $d_m = 128$, increasing the number of heads M leads to a gradual drop in performance. However, when d_m increases, the model has more ability to resist the lack of representation brought about by multi-head division. This is because each head has more learnable dimensional features for representation. When $d_m = 512$, increasing M brings positive gains. In contrast, the low feature dimension has limited representation and forcibly dividing it renders each part powerless. In contrast, when $d_m = 512$, a larger number of heads is required to prevent overfitting. Fig. 13(b) shows similar trends in counting tasks, and $d_m = 384$ yields the best performance with diverse indicators.

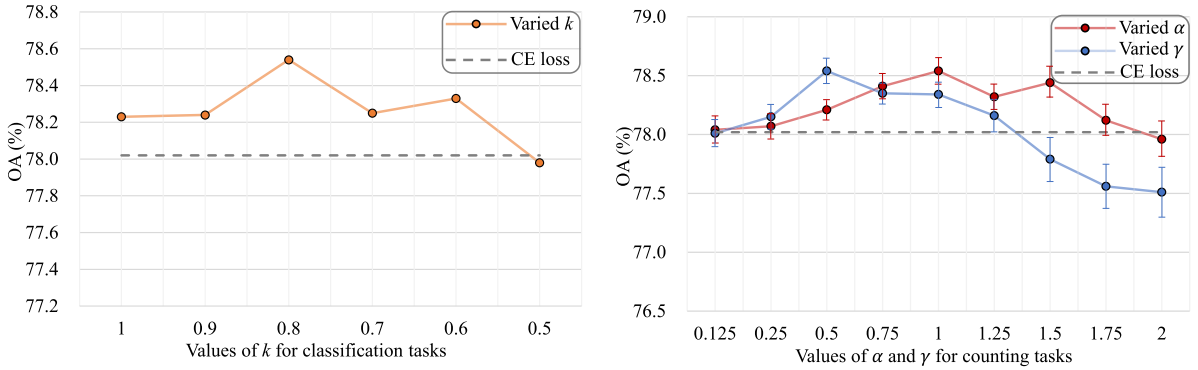
5.5. Adaptive numerical optimization

Because the proposed adaptive numerical optimization introduces three hyperparameters. k controls the intensity of hard example mining for classification tasks, while α and γ control the distance difference



(a) Varied α with $\gamma = 0.5$. (b) Varied γ with $\alpha = 1.0$.

Fig. 14. Relations between penalty d and distance difference $\mathbf{y}_{diff}$ with the controlled factors α and γ.



(a) OA with varied keep ratio k. (b) OA with varied α and γ.

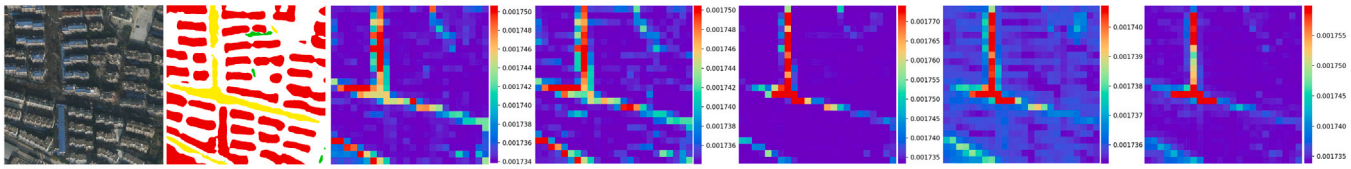
Fig. 15. Experimental results with varied hyperparameters for adaptive numerical optimization. The proposed adaptive numerical optimization outperforms the CE loss with a wide range of hyperparameter selections. (b) The mean values and standard deviations are reported after three runs.

penalty for counting tasks. Fig. 15(a) analyzes the optimization sensitivity with varied k. This shows that when $k > 0.6$, our balance strategy on classification optimization stably exceeds CE loss. The overall performance firstly increases and then slightly decreases from 1.0 to 0.5. This is because a low keep ratio reduces the number of effective samples in each batch, leading to model underfitting. By setting $k = 0.8$, we also varied α and γ from 0.125 to 2 and reported the overall accuracies in Fig. 15(b). We run each setting three times and mean value as well as standard deviation are reported. The results show that the proposed adaptive numerical optimization outperforms the CE loss with a wide range of hyperparameter selections, i.e., $\alpha \in (0, 1.75]$ and $\gamma \in (0, 1.25]$. To ensure stability in the VQA performance, it is recommended to restrict the value of γ into (0, 1]. The performance shows more instability when $\gamma > 1$, due to the influence curve changing from concave to convex (Fig. 14). Overall, setting these hyperparameters reasonably can lead to stable improvements in VQA performance.

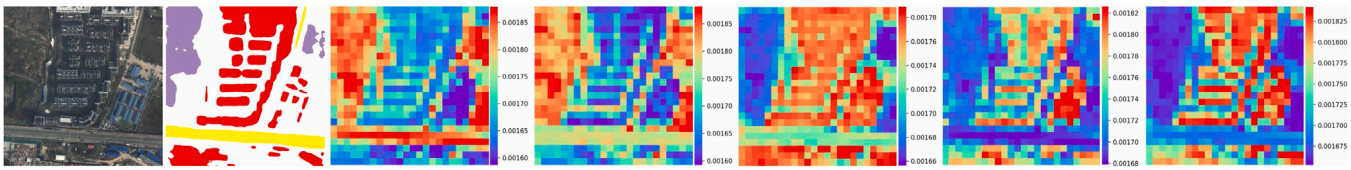
Additionally, we compare the proposed adaptive numerical optimization with two existing methods that are designed to address the sample imbalance problem, namely Focal Loss (Lin et al., 2017), CB Loss (Cui et al., 2019) and OHEM (Shrivastava et al., 2016). Table 6 shows that focal loss slightly improves the overall accuracy but reduces the counting accuracy. Although focal loss has balanced the hard-easy samples, it cannot adapt to the regression task. The CB loss considers both sample difficulty as well as class distribution and, also enhance the classification performance (OA+ = 0.18%). The uncertain number of optimized samples in each batch brings unstable training to OHEM, leading to negative effects for VQA tasks. Adding constraints to the classification and regression tasks separately results in different improvements. By combining these constraints, the proposed EarthVQANet achieves the best performance.

5.6. Overall architecture analysis

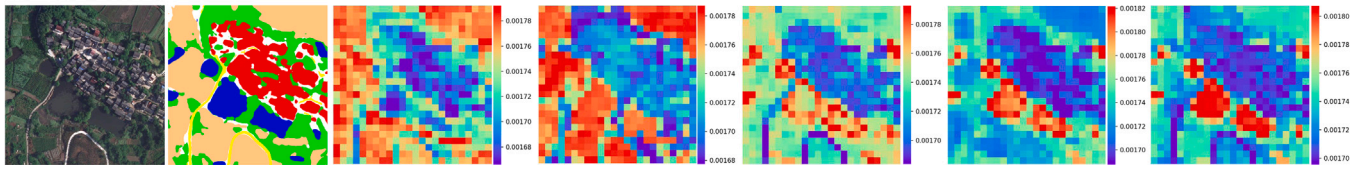
To comprehensively evaluate the contributions of each proposed module, comprehensive analysis experiments are performed. EarthVQANet was disassembled into five sub-modules, which are: (1) self-attention, (2) bidirectional cross-attention, (3) multi-task strategy, (4) semantic guidance, and 5) adaptive numerical optimization. Table 7 shows that each module contributes to the performance improvements in different ways. Bidirectional cross-attention achieves higher accuracies compared to self-attention, which highlights the importance of interaction between multi-modal features. Multi-task training further



(a) How many intersections are in this scene?



(b) What are the needs for the renovation of residents?



(c) What are the water types in this scene?

Fig. 16. Visualization of attention maps in semantic-guided cross-attention layers with language features as queries. From left to right are the raw image, semantic prediction and visualized attention maps from l_1 to l_5. Three examples are queried by different keywords: 'intersections', 'residents', and 'water'. As the layers deepen, the model progressively focus on the right queried words.

Table 6
Compared with the similar optimization algorithms.

Optimization	Classification	Regression	↑ OA (%)	↓ OR
CE loss (Baseline)	✓	✓	78.02	0.8146
Focal Loss (Lin et al., 2017)	✓	✓	78.11	0.8202
CB loss (Cui et al., 2019)	✓	✓	78.20	0.8154
OHEM (Shrivastava et al., 2016)	✓	✓	77.67	0.8362
+ balance on cls.	✓		78.18	0.8189
+ penalty on reg.		✓	78.23	0.7923
EarthVQANet	✓	✓	**78.54**	**0.7980**

improves performance due to the mutual promotion of segmentation and VQA. The addition of semantic guidance benefits the difficult questions that require object relational reasoning and significantly improves counting accuracy. All the proposed modules are compatible with each other within the EarthVQANet framework. Except for the necessary attention blocks, EarthVQANet maintains a lightweight design and efficiently achieves information extraction and knowledge reasoning.

For the model efficiency, we have reported the theoretical and practical indices. The theoretical indices include the floating-point operations (FLOPs) as well as model parameters. The theoretical metric is sample per second (FPS) that denotes the model inference speed. We have reported the inference speed after 500 runs on one 24G RTX 4090. The efficiency comparative results arec show in Table 8. Without the segmentation part, RSVQA achieves the highest efficiency but the lowest accuracy. Equipped with the segmentation network, BUTD and MCAN are supervised by the multi-task training, decreasing the model efficiency. Compared with MCAN, the EarthVQANet w/o guidance brings +6.22M on model size and 1.03 GFLOPs on prediction complexity. The semantic guidance further brings additional 2.14M parameters and slightly slowed down model inference speed (−4.24 Sample/Second).

5.7. Attention visualizations for keywords

The attention maps in the semantic-guided cross-attention layers are visualized to analyze the mechanism of multi-modal interactions. Since the attention map size is 24 × 24, we use bilinear interpolation to scale it to the size of a raw image for better comparison. Fig. 16(a) shows an urban scene and 'intersections' is queried to associate the visual features. The shallow layers (l_1 and l_2) divert attention on all roads in this scene, including arterial roads and internal paths for residents. However, as the number of layers deepens, the model gradually narrows its focus to the intersection of main roads, and ultimately concentrates on the intersection. Similarly, 'residents' is selected as the second query word in Fig. 16(b). The attention map in l_1 wrongly focuses on the road and barren areas but gradually shifts the attention to the right buildings in l_4 and l_5. The third example is a rural scene, and 'water' is selected to query the visual features. Due to the similar spectral signatures of water and agriculture, l_2 divides its attention to both the water and vegetation areas. However, in l_3, the model starts to filter out the uninteresting agricultural areas and concentrates on the water areas. Overall, the reasoning in the semantic-guided cross-attention layers is a step-by-step process that utilizes keywords to search for visual clues, reason about spatial semantic relationships, and finally summarize the knowledge.

6. Evaluation on other VQA datasets

6.1. Experiments on the FloodNet dataset

In order to test the generalizability of the proposed EarthVQANet, we also perform comparative experiments on FloodNet dataset (Rahnemoonfar et al., 2021). The FloodNet consists of 2343 post-disaster images obtained by unmanned aerial vehicles during the aftermath of Hurricane Harvey. The dataset covers Ford Bend County, Texas and other directly impacted areas. Because the QA pairs for the validation and test set are not released, we have re-arranged the training

Table 7
Overall architecture analysis.

Self-Att.	Bi. Cross-Att.	Multi-task	Sem. Guid.	AN optim.	Params(M)	↑mIoU (%)	↑OA (%)	↓OR
✓					52.44	–	75.16	1.1287
	✓				56.90	–	76.77	0.9943
✓	✓				58.83	–	76.96	0.9918
✓	✓	✓			63.13	56.97	77.55	0.8651
✓	✓	✓	✓		65.64	57.28	78.02	0.8146
✓	✓	✓	✓	✓	**65.64**	**57.43**	**78.54**	**0.7980**

Table 8
Efficiency analysis and comparison.

	Segm.	VQA	Guidance	Params (M)	FLOPs (G)	FPS (Sample/Second)	mIoU (%)	OA (%)
Segm (ConvX-T)	✓			33.69	108.35	46.32	52.27	–
RSVQA		✓		38.32	59.54	71.29	–	70.95
BUTD	✓	✓		53.59	161.23	25.86	57.11	77.55
MCAN	✓	✓		56.91	163.98	22.45	57.07	77.53
EarthVQANet	✓	✓		63.13	165.01	23.41	57.15	78.11
EarthVQANet	✓	✓	✓	65.54	226.29	19.17	57.43	78.54

Table 9
Compared results with other VQA methods on FloodNet^{Val} set.

Method	↑mIoU (%)	↑Accuracy (%)				↑OA (%)	↓RMSE		↓OR
		Com. Ju	Road Si.	Sim. Co	Com. Co		Sim. Co	Com. Co	
Only Segm.	72.61	–	–	–	–	–	–	–	–
• *General methods*									
SAN (Yang et al., 2016)	–	98.25	**99.11**	32.00	31.15	78.94	2.5171	2.9927	2.7768
MAC (Hudson and Manning, 2018)	–	98.84	98.45	36.80	29.71	79.16	5.0812	2.3494	4.0213
BUTD (Anderson et al., 2018)	72.69	**99.41**	98.89	35.20	35.50	80.18	2.2396	2.4657	2.3609
BAN (Kim et al., 2018b)	71.94	98.83	98.45	38.40	38.40	80.74	2.0842	3.1021	2.6672
D-VQA (Wen et al., 2021b)	71.96	98.83	98.89	32.80	36.95	79.98	1.8654	2.4286	2.1792
MCAN (Yu et al., 2019)	73.07	98.25	98.67	39.20	38.40	80.85	1.9819	2.5036	2.2706
• *RS methods*									
RSVQA (Lobry et al., 2020)	–	98.25	98.23	38.40	34.78	79.95	3.4871	4.1090	3.8260
RSIVQA (Zheng et al., 2021)	–	97.67	98.67	40.80	30.43	79.72	2.5059	3.1450	2.8591
EarthVQANet	**75.07**	98.47	98.85	**48.03**	**45.71**	**83.22**	**1.5163**	**2.4885**	**2.0834**

The abbreviations are: Com Ju (Complex Judging), Road Si (Road Situation Analysis), Sim Co (Simple Counting), Com Co (Complex Counting), OA (Overall Accuracy), OR (Overall RMSE).

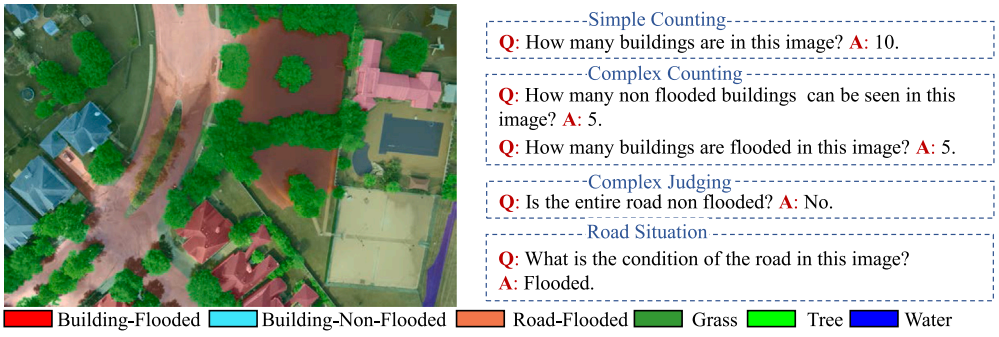


Fig. 17. Representative samples from the FloodNet dataset.

samples. Specifically, 1156 images and 3587 QA pairs are split for *Train* set and the rest are used for *Val* set. The original semantic categories include: background, building-flooded, building-non-flooded, road-flooded, road-no-flooded, vehicle, pool, tree, water and grass. The types of QA pairs are simple counting, complex judging, and object situation analysis.

For urban disaster experiments, we set the initial learning rate to 5e–5 and attention layer $N = 5$. Other settings remain unchanged. The comparative results are shown in Table 9. All the compared methods performed well on complex judging and road situation questions, as their candidate answers were simple and easy to identify (see Fig. 17). However, for complex counting questions, all methods obtained low counting accuracies, especially due to the wide variation in the number of attacked buildings in urban areas and the small-scale training samples. As a result, models trained on the FloodNet dataset are susceptible to overfitting. Consistent with the results from the EarthVQA dataset, models leveraging visual features demonstrated improved generalizability than global fusion methods. This further underscores the effectiveness of collaborative learning for segmentation and VQA tasks. Despite the limited samples and challenging disaster scenes, our proposed EarthVQANet method achieves the best overall performance, highlighting its potential to provide accurate VQA solutions in disaster scenarios (see Table 10).

Table 10
Compared results with other VQA methods on RSVQA-HR^{test}.

Method	Accuracy (%)				OA (%)
	Count	Presence	Compare	Area	
• *General methods*					
SAN (Yang et al., 2016)	68.42	91.47	90.77	85.25	84.37
MAC (Hudson and Manning, 2018)	68.08	90.94	90.68	85.57	84.09
BUTD (Anderson et al., 2018)	69.98	92.17	91.71	86.66	85.41
BAN (Kim et al., 2018b)	69.35	92.04	91.36	86.38	85.05
D-VQA (Wen et al., 2021b)	68.17	91.01	90.79	85.63	84.21
MCAN (Yu et al., 2019)	69.77	92.30	91.85	86.62	85.30
• *RS methods*					
RSVQA (Lobry et al., 2020)	69.03	91.35	91.04	83.22	83.77
RSVQA-R152[a] (Lobry et al., 2020)	68.63	90.43	88.19	85.24	83.23
RSIVQA (Zheng et al., 2021)	69.03	91.22	66.33	86.25	76.54
EarthVQANet	**70.75**	**92.45**	**92.43**	**87.10**	**85.98**

[a] The accuracies reported in original paper using ResNet152.

6.2. Experiments on the RSVQA dataset

As a traditional automatic generated VQA dataset, the RSVQA-HR dataset (Lobry et al., 2020) covers Portland, Manhattan and Philadelphia areas in America. Each 512 × 512 image has the resolution of 0.15 m which is extracted from the High Resolution Orthoimagery data collection of the United States Geological Survey. The whole dataset includes 10,659 images and 1,066,316 QA pairs, where training (61.5%), validation (11.2%), and test (27.3%) sets are splitted. As the RSVQA-HR dataset does not include the semantic masks, we convert all models into global feature fusion types. We trained all models using the Adam optimizer with a learning rate of 1e–5 for 35 epochs. The batch size is set to 16 and image augmentations include randomly vertical and horizontal flipping. As shown in Table 10, compared with the referenced VQA methods, the proposed EarthVQANet achieves the best accuracy on each type of question.

7. Conclusion

For the exploration of spatial and semantic relations between objects, the semantic segmentation and VQA tasks are collaboratively integrated for information extraction and knowledge reasoning. Specifically, the proposed EarthVQANet consists of a hierarchical pyramid segmentation network and semantic-guided attention. The hierarchical pyramid segmentation network is constructed for information extraction and provides accurate visual features as well as object semantics. The semantic-guided attention firstly utilizes self-attention to reason about object relations based on the guided visual features. The cross-attention is then designed to perform multi-modal interactions, searching for visual clues according to the questions and summarizing knowledge. Furthermore, we propose adaptive numerical optimization to unify the classification and regression tasks for VQA. Given an image and a question, EarthVQANet can automatically generate corresponding semantic maps and comprehensive answers. In experiments, EarthVQANet outperforms ten general and remote sensing VQA methods, including large multi-modal models (BLIP-2 and Instruct-BLIP). It also exhibits good scalability and friendly parameter sensitivity. We will extend the framework and dataset to other complex scenarios and applications.

CRediT authorship contribution statement

Junjue Wang: Conceptualization, Data curation, Formal analysis, Methodology, Validation, Writing – original draft. **Ailong Ma:** Conceptualization, Funding acquisition, Supervision, Writing – review & editing. **Zihang Chen:** Data curation, Visualization. **Zhuo Zheng:** Conceptualization, Investigation. **Yuting Wan:** Data curation, Validation, Visualization. **Liangpei Zhang:** Conceptualization, Supervision, Writing – review & editing. **Yanfei Zhong:** Conceptualization, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Natural Science Foundation of China under Grant 42325105 and 42171336.

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